Amr Zakariya

amrzakariya2018@gmail.com | +20 112 440 0492 | LinkedIn | GitHub: Amrzxk
Cairo, Egypt | Open to on-site, hybrid and remote

SUMMARY

Computer Science and AI graduate targeting junior DevOps and Cloud Engineering roles. Hands-on AWS across compute, serverless, networking, monitoring and IAM, with infrastructure defined as code in CloudFormation, SAM and Terraform, containerised workloads in Docker, and automated delivery through GitHub Actions. Comfortable on Linux, with practical experience in VPC design, cloud security and incident response.

EXPERIENCE

IT Intern · Genesis Creations S.A.E, Cairo *Oct 2025 – Jan 2026*

- Administered Microsoft 365 Admin Center across company accounts, configuring role-based access control, provisioning users, and managing access and licensing.
- Handled first-line incident response for hardware, software and network faults, diagnosing and resolving issues to minimise operational downtime.
- Maintained IT asset inventory and device lifecycle records, tracking assignments, accessories and employee handovers across the organisation.
- Worked alongside the engineering team on cloud and deployment operations, following release, automation and infrastructure practices in a live environment.

PROJECTS

Serverless URL Shortener · Personal Project *Jun 2025 – Jul 2025*

- Built a low-cost serverless redirection service on API Gateway, Lambda, DynamoDB, S3 and CloudFront.
- Defined 25+ resources as code with AWS SAM and CloudFormation, cutting full deployment time to under two minutes.
- Implemented keyless CI/CD in GitHub Actions using OIDC federation to short-lived IAM roles, eliminating long-lived credentials from the pipeline.
- Instrumented CloudWatch alarms, SNS notifications and AWS Budgets for early failure and cost-anomaly detection.
- Applied least-privilege IAM policies, blocked public S3 access and restricted CORS to enforce a minimal attack surface.

AWS Infrastructure Automation · NTI Program *Apr 2024 – Oct 2024*

- Designed a segmented VPC with public and private subnets, security groups and routing to isolate and protect application resources.
- Deployed a highly available web application on EC2 with static assets served from S3, sustaining 99.9% uptime.
- Configured CloudWatch monitoring and alerting to surface failures early and reduce incident response time.
- Automated backup and disaster recovery procedures, validating failover with minimal downtime under test.
- Enforced HTTPS using TLS certificates and redirection to strengthen transport security.

Hybrid Intrusion Detection System (H-IDS) · Graduation Project *Oct 2025 – Jun 2026*

- Engineered a real-time distributed architecture of three Redis Pub/Sub services streaming NFStream-captured traffic verdicts to a React dashboard over Server-Sent Events.
- Containerised the full stack with Docker Compose, running FastAPI under Gunicorn against PostgreSQL 16 with Alembic migrations.
- Paired a three-tier machine learning detection pipeline with Snort 3 signatures, sustaining roughly 111K flows per second.
- Secured the platform with argon2id hashing, JWT session cookies and role-based access control, and authored AWS EC2 deployment runbooks.

Kubernetes GitOps Lab · Personal Project, in progress *Aug 2026 – Present*

- Provisioning cluster infrastructure on AWS with Terraform, defining VPC, EC2 nodes, security groups and networking as code.
- Running a single-node k3s cluster hosting a containerised application deployed through declarative manifests.
- Implementing GitOps delivery with ArgoCD so cluster state is reconciled directly from a Git repository.

CERTIFICATIONS

In Progress: AWS Certified Solutions Architect – Associate (preparing via Manara, AWS cloud training partner for MENA)
AWS Academy: Cloud Architecting · Cloud Developer · Cloud Foundations
AWS (DEPI / NTI): Cloud Solution Administrator and Architect
Huawei Cloud: HCCDA Tech Essentials · HCCDP Cloud Migration
IBM: Containers and Kubernetes Essentials · Cloud Security

TECHNICAL SKILLS

Cloud: AWS (EC2, S3, Lambda, API Gateway, CloudFront, DynamoDB, VPC, IAM, CloudWatch, SNS, Budgets), Huawei Cloud
Infrastructure as Code: Terraform, AWS CloudFormation, AWS SAM
Containers and Orchestration: Docker, Docker Compose, Kubernetes, k3s
CI/CD and Version Control: GitHub Actions, OIDC federation, Git, GitHub
Monitoring and Observability: Amazon CloudWatch, Prometheus, Grafana
Networking and Security: TCP/IP, DNS, VPC design, security groups, IAM least-privilege, TLS/HTTPS, Wireshark
Systems and Data: Linux, Windows, Bash, Python, PostgreSQL, Redis, MySQL, SQL Server
Languages: Arabic (native), English (professional working proficiency)

EDUCATION

B.Sc. Computer Science and Artificial Intelligence · Capital University, Cairo *Oct 2022 – Jun 2026*
Computer Science track.